

DETERMINING THE ORDER OF MOULT OF THE BLACKSHOULDERED KITE *ELANUS CAERULEUS*

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Summary : The graphical display of skew-symmetry, using a
method due to Gower, is applied to moult data of the
Blackshouldered Kite *Elanus caeruleus*. From the
display, the general order in which the major flight
feathers moult is determined.

1. Introduction

Moult forms an important component of the annual cycle of birds, along with breeding and, for many species, migration between breeding and non-breeding areas. Statisticians have not given much attention to methods for the analysis of data gathered in moult studies (Ginn & Melville 1983). Underhill & Zucchini (in press) have considered the analysis of one type of moult data, that of primary moult scores, and have developed a model to estimate the parameters of primary moult. In this paper, a more general problem is considered, that of determining the order in which all the major flight feathers in the wing and tail are moulted, with more mathematical detail than that given in Underhill (1986).

Determining this order would be trivial if birds could be placed under

continuous observation. However, observations on birds in captivity cannot be extrapolated to wild populations. Nor is it possible to trap wild birds sufficiently frequently to enable order of moult to be determined. A further complication is that, in many species, there is no unique order of moult and there is considerable variation between individuals. The available data are thus a set of single observations on a sample of birds taken at various stages through the moult period. Each observation consists of a classification of each feather of interest into one of two categories – old (not yet moulted) or new (moulted). The aim of the analysis is to use these data to determine the order in which the feathers most frequently moult.

2. Data

The data used to illustrate the method consist of observations made on the moult of the ten primaries (labelled P1 to P10), thirteen secondaries (S1 to S13) and six rectrices (R1 to R6) on the wings and tails of Blackshouldered Kites *Elanus caeruleus* (figure 1, table 1). Birds were trapped throughout the moult period (mainly September to April) at a fairly uniform rate, but somewhat larger samples were obtained in October and December (table 2, Mendelsohn 1981).

The pattern for moult in Blackshouldered Kites is typical of many bird species: feathers are dropped one by one over a period of several months, each feather being replaced by a new feather within a few weeks, so that at any one time a small number of feathers are growing. As in most bird species, newly grown feathers are readily distinguishable from old feathers of the moult of the previous year by colouration and wear (Mendelsohn 1981, Ginn & Melville 1983).

Element a_{ij} of the 29×29 matrix A (table 1) indicates the number of times feather i was observed to moult after feather j . For example, primary 10 was observed to moult after primary 9 on 20 birds, but was never observed to moult before it; secondary 1 moulted after secondary 5 22

times, and before it nine times.

From table 1, the ordering of moult in the primaries is unambiguous. Without exception, primary 1 (the innermost primary, figure 1) moulted first, followed by primary 2, primary 3, ..., primary 10 (the outermost primary). In contrast, the pattern of moult for the secondaries and rectrices is not at all clear, and varies considerably from bird to bird. An extreme example is rectrix 1 which was observed to moult before, during and after the primary moult (table 1). Nevertheless, there is some pattern. In the secondaries, secondaries 5 and 13 were usually amongst the first to moult, whereas secondaries 8, 9 and 10 were generally the last to moult.

3. Graphical display techniques

Several graphical display techniques appear to be suitable for the data matrix (table 1). Firstly, it is clearly a contingency table, the leading case for the application of correspondence analysis (Greenacre 1984). Secondly, multidimensional scaling seems appropriate, since the elements may be treated as dissimilarities. This is motivated by the fact that if the moult of feathers i and j is widely separated in time, then they will frequently be observed as an old-new pair, and either a_{ij} or a_{ji} (or both) will be large.

Their average $m_{ij} = (a_{ij} + a_{ji})/2$ provides a measure of the "distance" between feathers i and j , so that the symmetric matrix $\mathbf{M}$ may be used as input to multidimensional scaling. This averaging method for handling asymmetry in dissimilarity data was suggested by Kruskal (1964).

Both these methods fail to produce a useful graphical display. Correspondence analysis focuses on the profiles of the rows and columns, which normally relate to the categories of two variables. However, it is meaningless to examine the profiles of row 1 and column 1 separately as if they were simply the first categories of two different variables. The fact that they both relate to primary 10 is not taken into account by correspondence analysis. Multidimensional scaling fails because the

symmetrisation process averages out the crucial information in the data, the information that one feather was moulted before (or after) another. In fact, the skew-symmetric component N of A contains the essential data: $n_{ij} = (a_{ij} - a_{ji})/2$. Note that $A = M + N$, that $\|A\|^2 = \|M\|^2 + \|N\|^2$, and that $\|N\|^2/\|A\|^2 = 0.665$, indicating that the skew-symmetric matrix N contains, by this measure, two-thirds of the information in A .

A graphical display technique for skew-symmetric matrices has recently been developed (Gower 1977, Constantine & Gower 1978), and has been applied to studies of emigration and immigration (Constantine & Gower 1982). The technique, which will be referred to as the Gower analysis, makes use of the special form which the singular value decomposition takes on for skew-symmetric matrices of order n :

$$N = U D_{\alpha} J U^t,$$

where the columns of U , the matrix of left singular vectors, are orthogonal, $D_{\alpha} = \text{diag}(\alpha_1, \alpha_2, \dots, \alpha_n)$, the diagonal matrix of ordered singular values with pairs of equal singular values, $\alpha_1 = \alpha_2$, $\alpha_3 = \alpha_4, \dots$ and J is an orthogonal matrix consisting of diagonal blocks

$$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

(If the order on N is odd, the final singular value in D_{α} is zero and the last diagonal element in J is one.) Since the singular values of skew-symmetric matrices have pairs of equal values, it only makes sense to consider matrix approximations in an even number of dimensions, say $2p$ (Gower 1977, 1984, Constantine & Gower 1978). We speak then of the matrix approximation in p "bimensions", a term coined by Harshman (1982). We confine ourselves to one bimension, and consider the rank two approximation of N , given by

$$N_{[2]} = \alpha_1(u_1 u_2^t - u_2 u_1^t),$$

where u_1 and u_2 are the first two columns of U , associated with the

singular value α_1 . By the Eckart-Young theorem (1936), $N_{[2]}$ gives the best least-squares fit to N of all rank two matrices, and the quality of the approximation is $100\alpha_1/\sum_{i=1}^{[n/2]}\alpha_{2i}$ per cent.

To form the graphical display of the first bimension, one plots the n points $(u_{1i}u_{2i})$.

By an elementary property of determinants, the area of the triangle formed by the origin and points i and j (ΔO_{ij}) is a half of $(u_{1i}u_{2j} - u_{1j}u_{2i})$ which, in turn, is equal to $n_{[2]ij}/\alpha_1$. Thus, if the rank two approximation is good,

$$n_{ij} \approx n_{[2]ij} = \frac{\alpha_1}{2} \Delta O_{ij},$$

and the skew-symmetry is accommodated since $\Delta O_{ij} = -\Delta O_{ji}$.

We have the following guidelines for the interpretation of this unusual space (Gower 1978):

1. Consider the j -th column of N and draw a line through point j and the origin. All those points i for which n_{ij} is positive ought to lie in one of the half-planes defined by the line, and all points i for which n_{ij} is negative to lie in the opposite half-plane.
2. As n_{ij} increases, point i should lie at an increasing perpendicular distance from the line through point j and the origin.
3. If $n_{ij} = n_{kj}$ then $\Delta O_{ij} \approx \Delta O_{kj}$. Since these triangles have the same base, their heights must be nearly equal, implying that points i and k must lie approximately on a line parallel to the line through j and the origin.

A special form of skew-symmetry arises when all the points in the plot lie on a straight line. Then $N = \alpha_I(uI^t - Iu^t)$, so that $n_{ij} = \alpha_I(u_i - u_j)$. All the triangles in the plot then have the same height, and the skew-symmetries are proportional to the differences $u_i - u_j$ in the bases.

4. Results

The plot resulting from the application of the Gower analysis to the skew-symmetric matrix $N = A - A^t$ (figure 2) has quality 99.7%. It shows the points for the feathers lying very close to a straight line. This suggests that the special simple form of skew-symmetry, $n_{ij} = \alpha(u_i - u_j)$, is appropriate for these data. The line of points in figure 1 also suggests that they may be used to form an ordering for the moult of the 29 feathers. That figure 1 can be interpreted in this way is justified by considering that, if two feathers i and j generally moult at the same time, then $n_{ij} \approx 0$ so that $\Delta O_{ij} \approx 0$. This implies that points i and j are approximately coincident. If feather i moults before feather k , but after feather l , then n_{ik} and n_{il} have opposite signs, and the points representing feathers k and l lie on opposite sides of point i in the line of points. The orientation of the ordering needs to be determined from inspection. In this case it is known that primary 1 is the first primary to moult, followed by the other primaries in sequence. This simple ordering of primary moult is represented in figure 1. Using the notations $i \rightarrow j$ to indicate that feather i moults before feather j and $i \& j$ to indicate that feathers i and j moult simultaneously, the suggested ordering of secondary moult in figure 1 is

$5 \rightarrow 1 \& 13 \rightarrow 6 \& 12 \rightarrow 11 \rightarrow 2 \rightarrow 7 \rightarrow 4 \rightarrow 8 \rightarrow 3 \rightarrow 9 \& 10$.

The order of moult for the rectrices is

$1 \rightarrow 2 \rightarrow 3 \rightarrow 6 \rightarrow 4 \rightarrow 5$.

It is not claimed that all birds follow this order, but that this is the general order in the "average" bird.

If it may be assumed that the data were collected uniformly through the moult period, it seems reasonable to assume that figure 2 demonstrates something stronger than mere ordering of moult, and contains information about the relative timing. For example, it seems fair to conclude that primaries 1 and 2 moult in quick succession, with longer and nearly equal time periods between the moult of primaries 2 and 3, 3 and 4, ..., 6 and 7 (figure 2). One simple way of estimating the timing of moult would be to project the points in figure 2 onto straight line. A more sophisticated approach, is to use the robust median polish technique of Tukey (1977) to fit the additive model $n_{ij} = \alpha_i + \beta_j$ with $\beta_j = -\alpha_j$ (table 3). The fitting of this model is justified and motivated by the special simple form of the skew-symmetry, diagnosed by the plot of the Gower analysis. Rescaling the main effects linearly between 0 and 100 (table 3) simplifies the interpretation: primary 2 is moulted at 4% of the way through moult, secondary 1 at 61%, etc. A fuller ornithological analysis of the results may be found in Underhill (1986) and is not duplicated here.

5. Discussion

If feathers moulted at random, the rank two approximation would be poor, and the plot of the first dimension would show an uninterpretable scatter of points. The more constant the order of moult is, the closer, in general, the points should lie along a line. This presupposes that the sample of birds examined is a random sample of birds in moult.

Since all birds do not commence and end moult simultaneously, it is non-trivial to devise a sampling plan that will ensure a random sample. The best advice that can be offered is to attempt to capture birds as evenly as possible throughout the period during which birds in the population are moulting. Based on the experience obtained from the example considered, the method does not appear to be overly sensitive to fairly substantial departure from even sampling (table 2).

Even if the sample of birds is a random sample of birds in all stages of moult, the interpretation that distance between points representing feathers is proportional to time between their moult needs to be treated with some caution. Suppose that the times of moult of feathers I and J are independent random variables on the interval $(0, T)$ with distribution functions $F_I(t)$ and $F_J(t)$, respectively. Then, at time t , one of four events may be observed: neither I nor J has moulted, with probability $(1-F_I(t))(1-F_J(t))$, or both I and J have moulted, with probability $F_I(t)F_J(t)$, or I has not moulted but J has, with probability $(1-F_I(t))F_J(t)$, or I has moulted but not J , with probability $F_I(t)(1-F_J(t))$. The last two events are related to the observations α_{IJ} and α_{JI} in the data matrix. The difference between the probabilities of these two events, integrated over the period of moult, is proportional to n_{IJ} :

$$n_{IJ} \propto \int_0^T F_I(t) - F_J(t) dt.$$

If $F_I(t)$ and $F_J(t)$ have the same shape, differing only in mean time of moult μ_I and μ_J respectively, this integral is approximately $\mu_I - \mu_J$, and the interpretation is justified. However, if $F_I(t)$ and $F_J(t)$ differ in shape, so that for example, their means and variances are $\mu_I < \mu_J$ and $\sigma_I^2 < \sigma_J^2$, and $F_I(t) > F_J(t)$ for some interval of values t , then

$$\int_0^T F_I(t) - F_J(t) dt < \mu_I - \mu_J,$$

so that points I and J in the graphical display will tend to be closer together than they ought to be. A positive correlation between the times of moult of feathers I and J in a single bird is more likely than independence, and this will tend to reduce this shrinkage effect.

In the example considered, the timing of moult of the rectrices was very

variable in comparison with the primaries. We thus expect shrinkage in the plot of the distances between points for primaries and rectrices, but no shrinkage in the inter-primary distances. These are conflicting requirements, and the plot represents a compromise between them. In conclusion, it seems likely that errors introduced into figure 2 through uneven sampling are greater than shrinkage errors and that this plot may safely be interpreted to give rank ordering of moult. However, interpretation of timing of moult, as given in table 3, needs to be treated with some reserve. A simulation study could be designed to provide guidelines to the extent to which interval information in the plot can be interpreted.

The Gower analysis of skew-symmetry appears to be poorly known, but should always be considered as a useful exploratory tool for gaining insight into data in the form of an asymmetric square matrix. Some examples include brand switching (e.g. frequency with which motorists bought car of model i last time and model j this time), confusion matrices (e.g. number of times an advertisement for product i is thought to be for product j), the frequency with which journal i is cited by journal j , trade and migration between countries, the number of people who live in zone i but work in zone j , the number of marriages in which the bride is age i and the groom is age j , and input-output matrices in economics.

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TABLE 1. The numbers of Blackshouldered Kites trapped and in moult in each month

	Number trapped	Number in moult
July	22	4
August	16	2
September	24	17
October	43	33
November	14	14
December	31	31
January	10	10
February	20	20
March	21	19
April	12	6
May	8	3
June	12	2

TABLE 2. Moult data for the Blackshouldered Kite (from Mendelsohn 1981), showing the number of times the feather in the row was observed to moult after the feather in the column. Primaries 1 to 10 are labelled P1 to P10 and numbered outwards. The secondaries S1 to S13 are numbered inwards and the rectrices R1 to R6 are numbered from the centre outwards.

	P10	P9	P8	P7	P6	P5	P4	P3	P2	P1	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12	S13	R6	R5	R4	R3	R2	R1
P10	0	20	30	44	61	79	96	114	131	135	54	42	30	33	36	53	36	30	25	27	45	52	56	41	28	41	54	67	95
P9	0	0	21	35	53	70	87	105	122	126	45	32	20	24	53	44	26	19	15	18	36	44	46	39	24	32	46	60	87
P8	0	0	0	20	38	55	72	90	107	111	31	18	6	12	39	29	11	6	4	6	22	32	35	21	6	18	34	47	75
P7	0	0	0	0	31	48	65	83	100	104	24	6	4	9	34	20	4	4	3	4	14	26	30	6	3	14	26	39	66
P6	0	0	0	0	0	35	52	70	87	91	8	4	4	7	24	8	2	2	1	3	7	14	20	2	1	7	14	27	54
P5	0	0	0	0	0	0	37	55	72	76	4	3	3	3	10	4	1	1	0	2	3	5	15	0	0	4	4	15	43
P4	0	0	0	0	0	0	0	38	54	59	0	2	1	1	5	1	1	1	0	1	1	2	7	0	0	3	3	7	26
P3	0	0	0	0	0	0	0	0	35	39	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	2	1	1	19
P2	0	0	0	0	0	0	0	0	0	22	0	1	1	0	1	1	1	0	0	0	0	0	1	0	1	2	1	1	10
P1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	7
S1	0	0	0	3	15	34	52	69	86	90	0	1	0	5	22	7	0	1	1	1	5	15	19	6	1	8	14	27	55
S2	0	0	2	14	32	49	66	82	99	103	24	0	3	8	33	22	3	4	3	1	13	28	30	18	5	15	27	40	68
S3	1	4	9	24	40	56	74	91	108	112	32	18	0	11	41	29	14	8	6	7	24	34	38	23	11	20	35	48	76
S4	0	4	8	21	36	51	70	87	104	108	30	17	8	0	37	26	13	8	3	7	19	30	34	21	8	18	32	45	70
S5	0	1	3	8	13	25	43	60	78	82	9	6	4	5	0	6	3	3	2	2	7	9	17	5	3	11	11	23	48
S6	0	1	2	7	20	39	56	74	91	95	15	4	3	6	23	0	1	2	1	1	6	18	23	9	0	9	19	34	59
S7	0	0	2	16	25	51	68	85	102	106	25	10	4	9	35	24	0	3	3	2	13	28	30	20	4	14	29	43	69
S8	0	1	7	20	38	54	71	88	105	109	29	17	6	9	38	27	12	0	1	6	21	30	33	20	8	19	33	45	72
S9	0	3	9	22	40	56	73	91	108	112	32	20	11	12	40	30	15	7	0	8	23	32	37	22	9	20	35	49	74
S10	1	3	9	25	41	59	76	93	110	114	35	22	9	14	43	32	16	8	7	0	25	35	37	26	12	23	38	50	77
S11	0	2	3	14	29	44	62	79	96	100	21	9	6	9	30	19	3	5	4	1	0	23	24	18	4	13	27	37	63
S12	2	5	7	12	22	36	56	75	91	96	17	10	9	12	25	12	7	7	5	8	10	0	22	10	5	15	22	30	55
S13	0	2	8	14	23	36	51	68	86	90	19	11	8	10	22	17	8	8	7	8	9	11	0	11	7	14	20	29	53
R6	2	4	6	14	34	50	68	86	102	107	28	14	9	11	35	25	11	8	5	9	17	24	31	0	5	13	26	36	66
R5	1	2	10	24	42	58	75	93	110	114	34	21	11	15	42	32	15	10	7	10	25	34	36	23	0	20	36	49	75
R4	3	5	8	19	37	53	71	88	105	110	30	19	11	13	39	29	13	11	8	9	22	31	33	19	4	0	30	46	68
R3	2	5	6	10	21	38	56	74	92	95	19	9	8	11	29	17	8	8	6	8	14	20	25	5	5	12	0	31	59
R2	3	5	6	13	20	35	52	73	92	96	19	11	6	9	25	16	10	6	5	8	16	21	21	9	7	10	21	0	59
R1	4	4	8	10	18	26	41	61	71	7	9	15	10	9	10	21	13	11	7	5	9	12	16	9	6	9	13	18	0

TABLE 3. Results of application of Tukey's (1977) median polish to the data of table 2. In the final column, the main effects have been rescaled to lie between 0 and 100.

feather	main effects	rescaled
P10	22,0	100
P9	16,5	93
P8	8,0	81
P7	1,5	72
P6	-9,5	57
P5	-19,5	43
P4	-29,5	29
P3	-39,5	16
P2	-48,0	4
P1	-51,0	0
S1	-6,5	61
S2	2,0	73
S3	8,5	82
S4	5,5	77
S5	-11,0	55
S6	-4,5	64
S7	4,0	64
S8	7,5	80
S9	9,5	83
S10	9,5	83
S11	0,0	70
S12	-4,0	64
S13	-6,5	61
R6	2,5	73
R5	9,5	83
R4	3,5	75
R3	-5,0	63
R2	-11,0	55
R1	-25,0	36

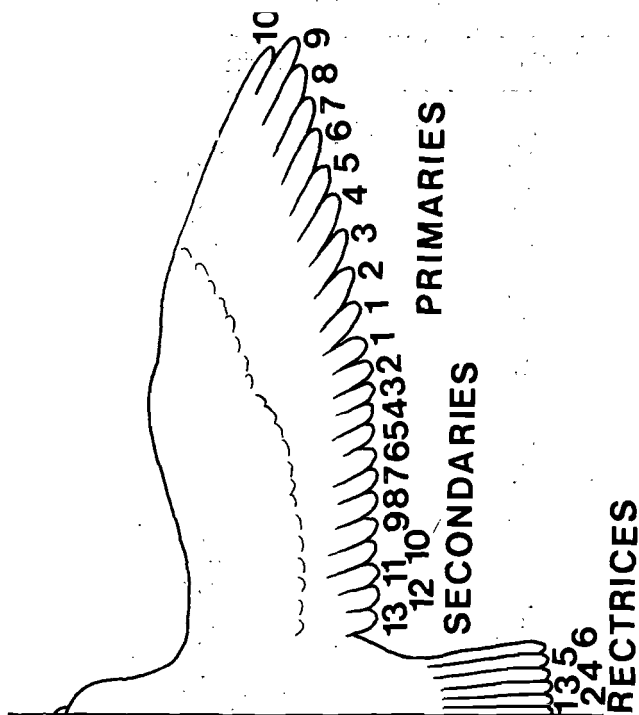


Figure 1. Diagram showing the numbering of the primaries, secondaries and tail of a bird.

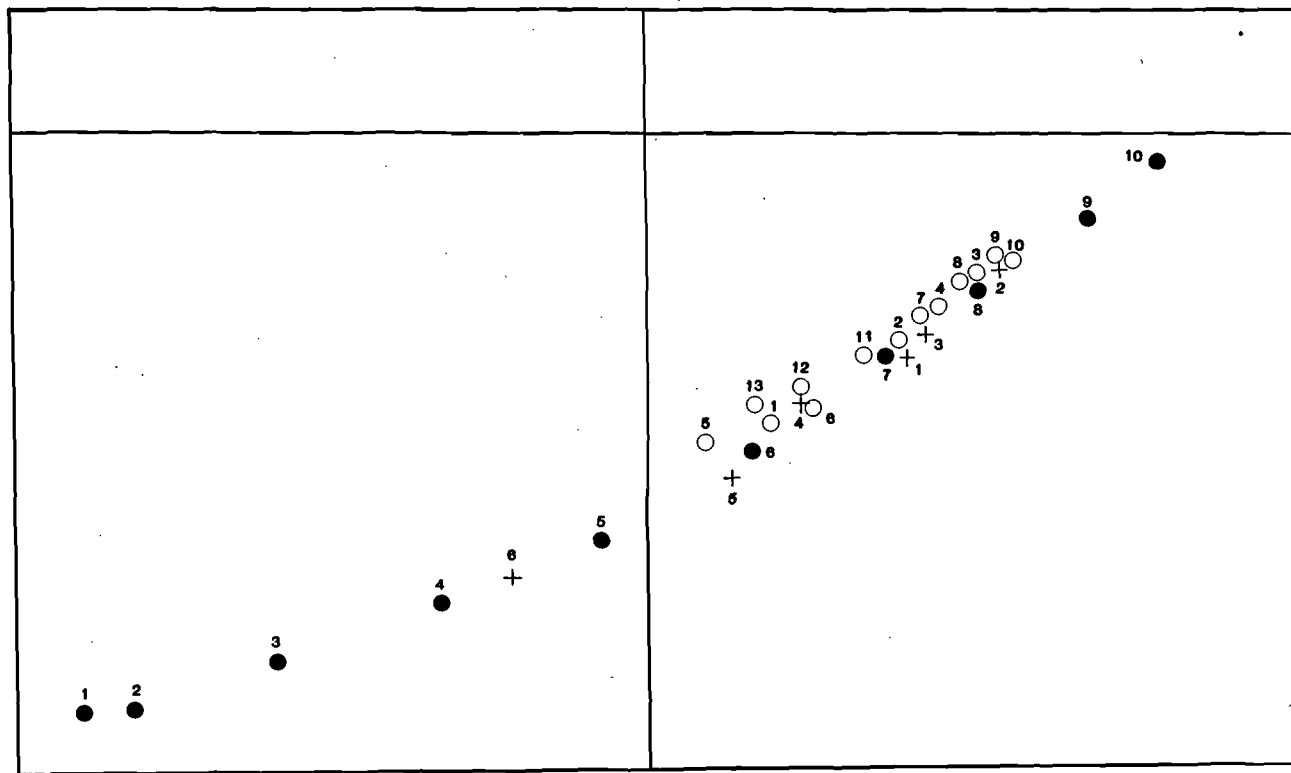


FIGURE 2. The plot of the skew symmetric matrix derived from table 2. The primaries (closed circles) are labelled 1 to 10, the secondaries 1 to 13, and the rectrices (crosses) 1 to 6. The magnitude of the triangle made by the points for two feathers and the origin in difference in the timing of moult for the feathers.